

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2026

PHYSICS PAPER 1

8:30 am – 11:00 am (2½ hours)

This paper must be answered in English

GENERAL INSTRUCTIONS

- (1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 50 minutes.
- (2) Section A consists of multiple-choice questions in this question paper, while Section B contains conventional questions printed separately in Question-Answer Book **B**.
- (3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in the Question-Answer Book. **The Answer Sheet for Section A and the Question-Answer Book for Section B will be collected separately at the end of the examination.**
- (4) The diagrams in this paper are **NOT** necessarily drawn to scale.
- (5) The last two pages of this question paper contain a list of data, formulae and relationships which you may find useful.

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

- (1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
- (2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.
- (3) All questions carry equal marks.
- (4) **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
- (5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
- (6) No marks will be deducted for wrong answers.

Section A

There are 33 questions. Questions marked with * involve knowledge of the extension component.
Take $g = 9.81 \text{ m s}^{-2}$.

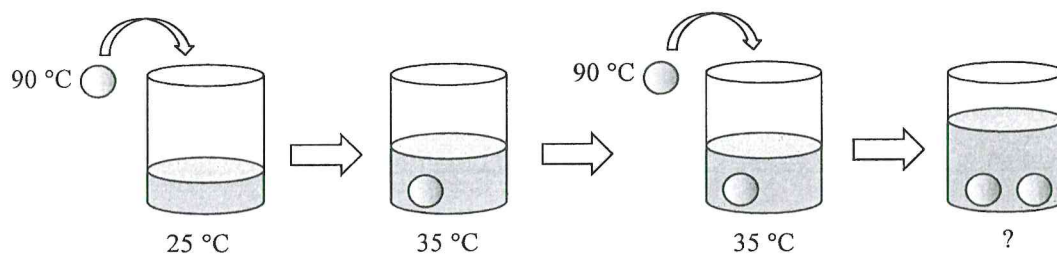
1. Flasks X and Y are identical except that for their outer surfaces: X is black and Y is silvery. Each flask contains the same amount of water initially at 25°C , sealed and equipped with a temperature sensor. The flasks are then placed in the same outdoor environment at 30°C with sunlight. After 20 minutes,

- A. the water in X is hotter as a black surface is a better absorber of radiation.
- B. the water in X is hotter as a black surface is a better emitter of radiation.
- C. the water in Y is hotter as a silvery surface is a poorer absorber of radiation.
- D. the water in Y is hotter as a silvery surface is a poorer emitter of radiation.

2. The resistance of a resistance thermometer is 100.0Ω in pure melting ice. Its resistance increases linearly with temperature by $0.385 \Omega ^\circ\text{C}^{-1}$. Find the temperature when the resistance is 117.2Ω .

- A. 6.62°C
- B. 11.5°C
- C. 44.7°C
- D. 45.1°C

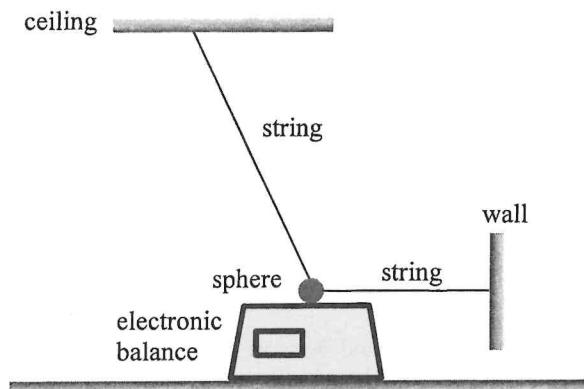
3.



A metal sphere at 90°C is put into a cup of water. The temperature of the water increases from 25°C to 35°C . If a second identical metal sphere, also at 90°C , is added to the water, what is the final temperature of the water? Assume there is no heat loss to the cup or the surroundings throughout the process.

- A. 41.5°C
- B. 42.3°C
- C. 43.5°C
- D. 45.0°C

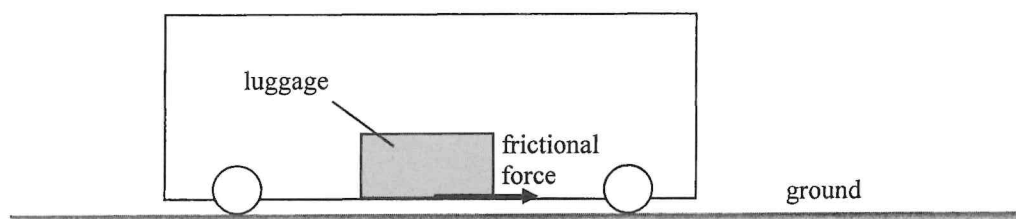
4.



A sphere of weight 18 N is suspended by two light and inextensible strings. One string is attached horizontally to a wall, and the other is connected to the ceiling. The sphere rests on the smooth top surface of an electronic balance, which shows a reading of 10 N . Find the resultant force acting on the sphere by the two strings.

- A. 0 N
- B. 8 N
- C. 10 N
- D. 15 N

5. A train is moving horizontally along a straight line. A piece of luggage rests on the floor of the train. The diagram shows the frictional force acting on the luggage, directed to the right. Neglect air resistance.

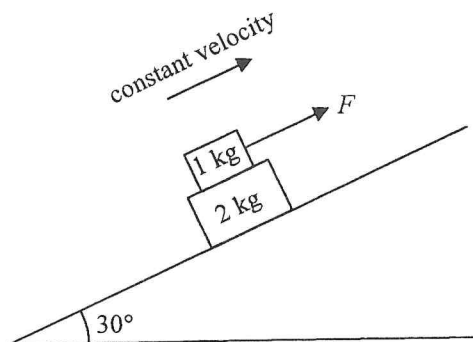


Observing from the ground, which of the following statements must be correct ?

- (1) The train is travelling to the right.
- (2) The train is accelerating.
- (3) The direction of acceleration of the luggage is to the right.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

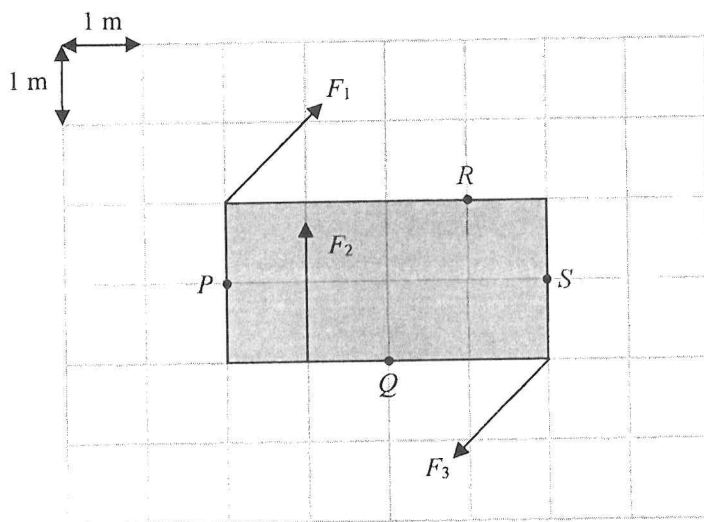
6.



A 1-kg block is placed on a 2-kg block, and both are on a smooth incline at 30° to the horizontal. A constant force F is applied to the 1-kg block upward along the incline so that the two blocks move up the incline together with the same constant velocity as shown. What is the frictional force acting on the 1-kg block by the 2-kg block? Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

- A. 4.91 N upward along the incline
- B. 9.81 N upward along the incline
- C. 4.91 N downward along the incline
- D. 9.81 N downward along the incline

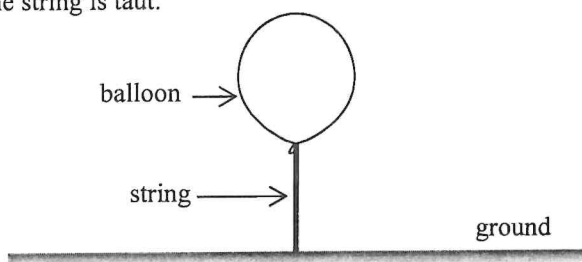
7.



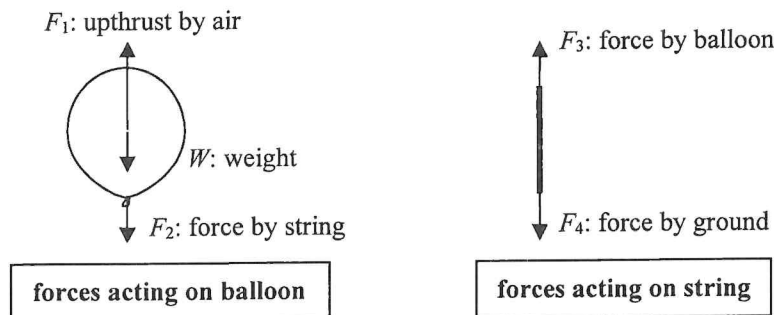
A rectangular plate of width 2 m and length 4 m is placed on a smooth horizontal surface. Three forces F_1 , F_2 and F_3 of equal magnitude act on it as shown. F_1 and F_3 are parallel and opposite in direction. Taking the moment about which point below is the net moment the greatest?

- A. P
- B. Q
- C. R
- D. S

8. One end of a light and inextensible string is connected to a floating balloon and the other end is fixed to the ground as shown. The string is taut.



The free-body diagrams of the balloon and the string are shown below.

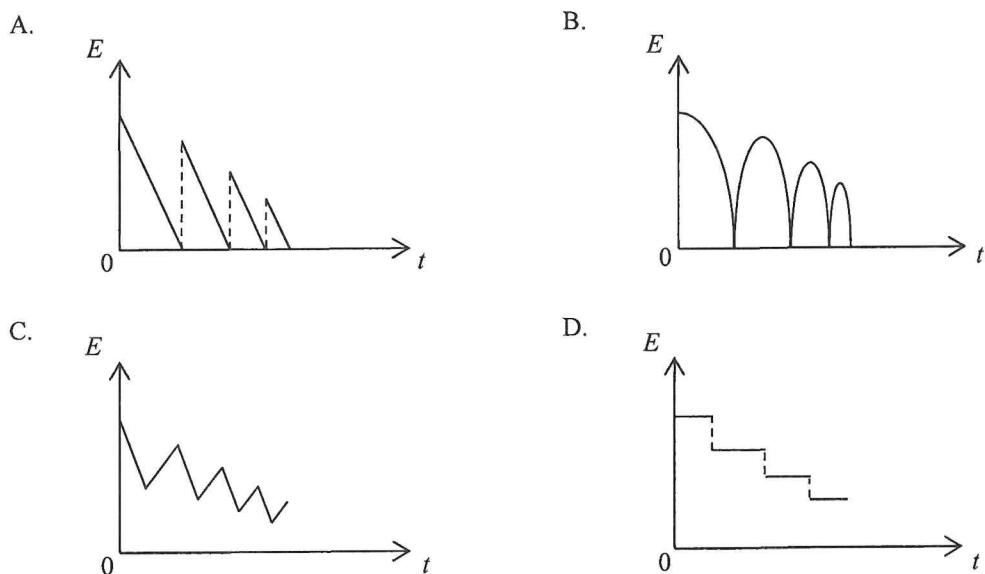


Which of the following is/are action and reaction pair(s) ?

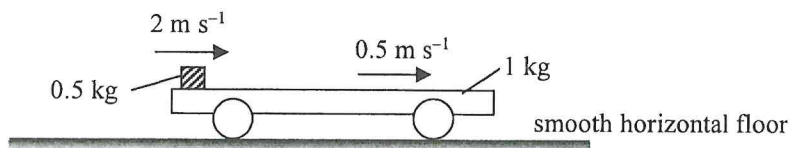
- (1) F_1 and W
- (2) F_2 and F_3
- (3) F_3 and F_4

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

9. A ball is released from a certain height. It rebounds from the ground several times before it comes to rest. Neglect air resistance. Which of the following graphs best represents the variation of the total mechanical energy E of the ball with time t ?



10. A 1-kg trolley with a 0.5-kg block placed on its horizontal platform is resting on a smooth horizontal floor. The block is then given a sharp push to the right. At an instant after the push, the block and the trolley are moving to the right at 2 m s^{-1} and 0.5 m s^{-1} respectively as observed from the floor.



Finally, they travel together with the same speed v . Find v . Neglect air resistance.

- A. 0.5 m s^{-1}
 B. 1 m s^{-1}
 C. 1.5 m s^{-1}
 D. 2 m s^{-1}
- *11. Satellites X and Y revolve around the Earth in circular orbits, in the same plane and in the same direction. The radii of their orbits are r and $4r$ respectively. The figure below shows an instant that X , Y and the Earth are on a straight line.

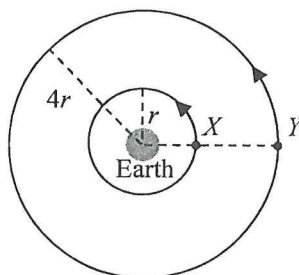
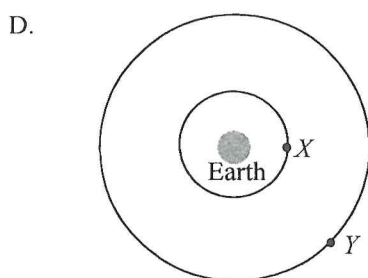
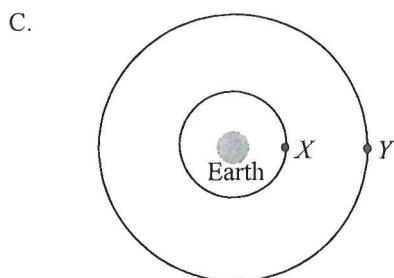
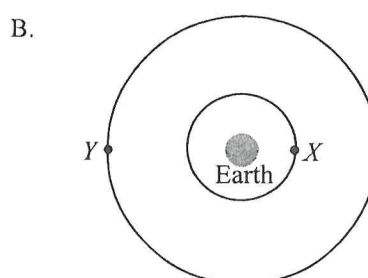
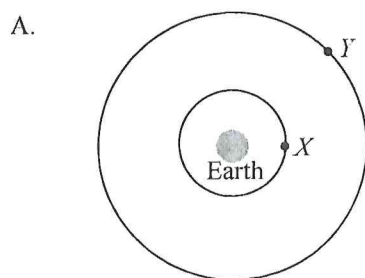


Diagram NOT drawn to scale

When X completes one cycle, which of the following diagrams best represents the position of Y ?



- *12. A rocket on the Earth's surface is launched by an upward thrust F , producing an upward acceleration g , where g is the acceleration due to gravity near the Earth's surface, as shown in Figure (a). R is the radius of the Earth.

Diagram NOT drawn to scale

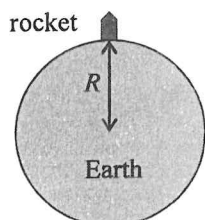


Figure (a)

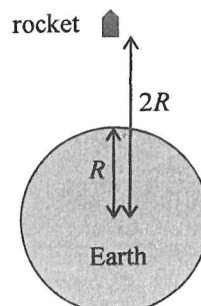
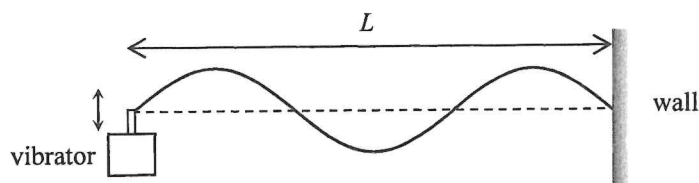


Figure (b)

Figure (b) shows an identical rocket at a distance $2R$ away from the Earth's centre. What is the upward thrust required for producing the same upward acceleration g ? Neglect air resistance.

- A. $\frac{F}{4}$
 B. $\frac{F}{2}$
 C. $\frac{5F}{8}$
 D. $\frac{3F}{4}$
13. The figure shows a string with one end fixed on the wall and the other end tied to a vibrator with frequency f . A stationary wave is formed as shown at a certain instant. The distance between the vibrator and the wall is L .



Which of the following statements are correct?

- (1) All particles on the string between two successive nodes are vibrating in phase.
 (2) If the tension in the string increases while L remains unchanged, f must increase to maintain the same stationary wave pattern.
 (3) The wavelength of the stationary wave is $\frac{L}{3}$.

- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

14. Which of the following phenomena involve(s) the principle of superposition of waves ?
- (1) Ripples in a pond overlap and create regions with different amplitudes.
 - (2) Stationary wave patterns are formed when two waves of the same frequency travel in opposite directions.
 - (3) The speed of waves changes when they travel from one medium to another.
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

15.

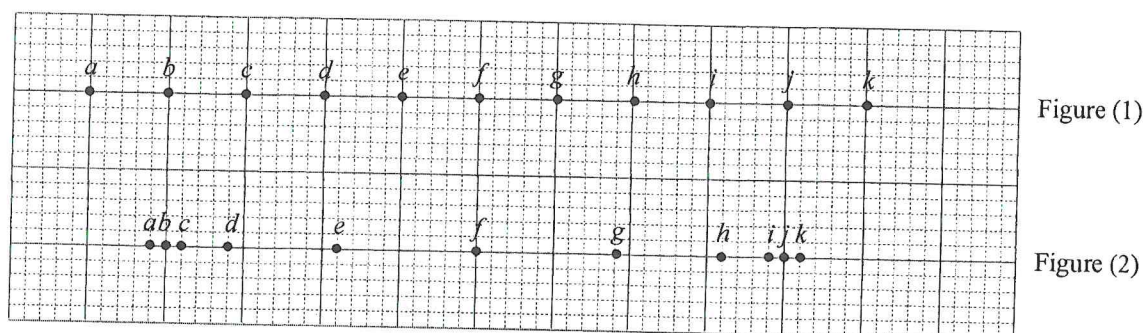
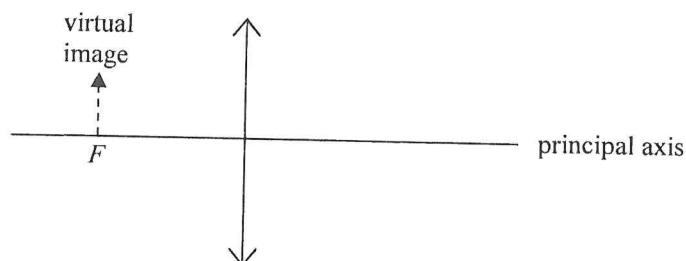


Figure (1) shows the equilibrium positions of particles *a* to *k* in a medium. A longitudinal wave is travelling through the medium and the positions of the particles at a certain instant are shown in Figure (2). Which of the following statements is/are correct at that instant ?

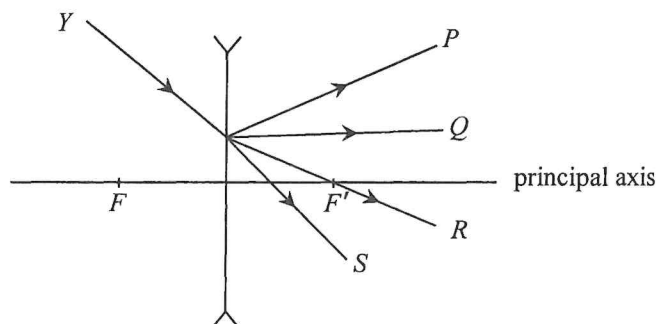
- (1) The separation between *c* and *k* is one wavelength.
 - (2) *f* is momentarily at rest.
 - (3) *i* and *k* are moving in the same direction.
- A. (2) only
 B. (3) only
 C. (1) and (2) only
 D. (1) and (3) only
16. The image of an object formed by a convex lens is virtual and is situated at the focus *F* as shown.



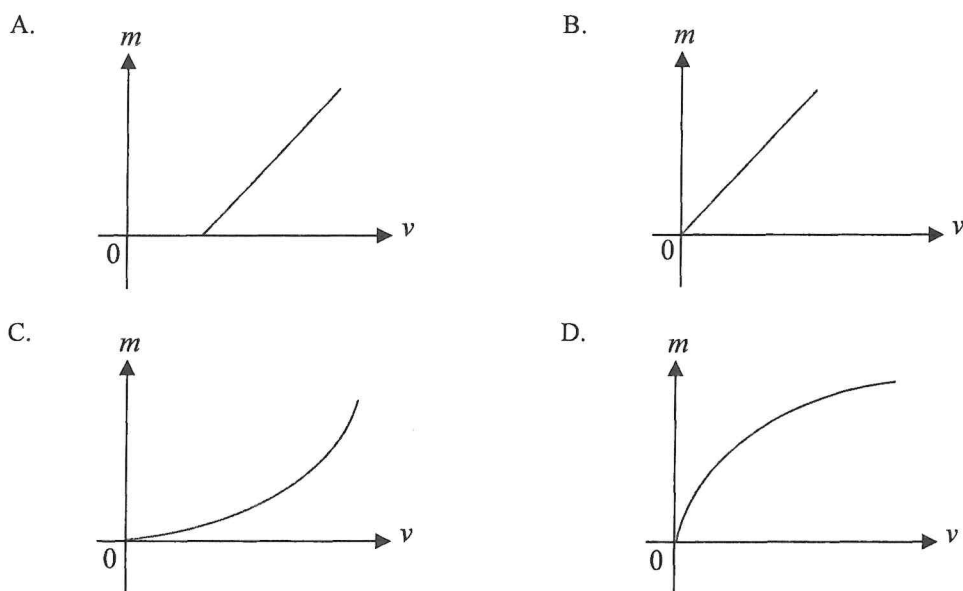
Which of the following statements is/are correct ?

- (1) The object is at infinity.
 - (2) The object is larger than the image.
 - (3) The object and the image are on the same side of the lens.
- A. (2) only
 B. (3) only
 C. (1) and (2) only
 D. (1) and (3) only

17. Light ray Y is incident on a concave lens as shown. F and F' are the foci of the lens. Which light ray best represents the emergent ray?



- A. P
 B. Q
 C. R
 D. S
- *18. A convex lens is used to project sharp images of an object onto a screen at different object distances u . For each value of u , the corresponding image distance v and the linear magnification m are recorded. Which of the following best represents the graph of m against v ?

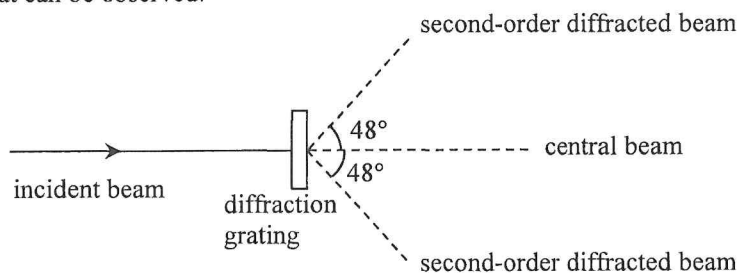


19. A monochromatic light beam is directed to a double slit and a fringe pattern is formed. Which of the following physical phenomena is/are involved?

- (1) dispersion
 (2) diffraction
 (3) interference

- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

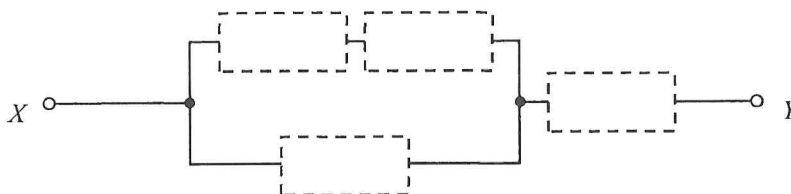
- *20. A beam of monochromatic light is incident normally on a diffraction grating. Second-order diffracted beams are found at 48° to the incident direction as shown in the figure. Find the highest order of diffracted beam that can be observed.



- A. 2nd order
 B. 3rd order
 C. 4th order
 D. 5th order
21. Two small charged spheres exert an electrostatic force of magnitude F on each other. If the amount of charge on each sphere is doubled and the distance between them is also doubled, the magnitude of the electrostatic force becomes

- A. $\frac{F}{2}$
 B. F
 C. $2F$
 D. $4F$

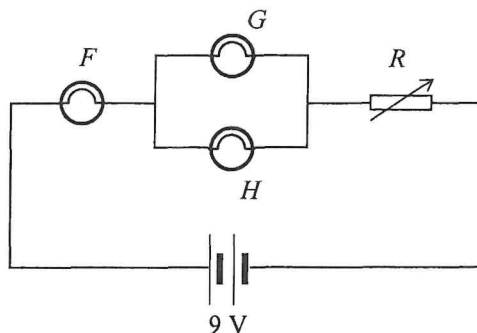
22.



Four resistors of resistance $10\ \Omega$, $20\ \Omega$, $30\ \Omega$ and $40\ \Omega$ respectively are installed in the empty places in the above circuit. What is the minimum possible equivalent resistance across X and Y ?

- A. $19.2\ \Omega$
 B. $25.6\ \Omega$
 C. $28.8\ \Omega$
 D. $32.2\ \Omega$

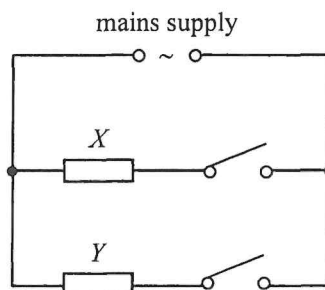
23.



In the circuit above, light bulbs F , G and H are identical and the battery is ideal. Rheostat R is initially set to maximum resistance. The maximum voltage across each bulb without burning out is 2 V, and a burnt-out bulb becomes an open circuit. Initially all bulbs light up. If the resistance of R is gradually reduced to zero, which of the bulbs will burn out in the process ?

- A. only bulb F
- B. only bulbs G and H
- C. none of them
- D. all of them

24.



The above shows a simplified diagram of the circuit in a heater. It consists of two heating elements X and Y . The time taken for the heater to heat a certain amount of water from room temperature to 100°C is as follows.

only X is connected	180 s
only Y is connected	300 s

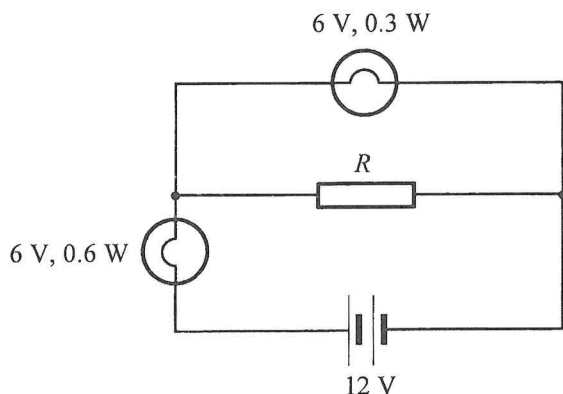
How long does it take to heat the same amount of water from room temperature to 100°C if both X and Y are connected ?

- A. 105 s
- B. 113 s
- C. 154 s
- D. 240 s

25. In household appliances, the function of the earth wire is

- A. to provide a return path for the current during normal operations.
- B. to provide a return path for the current when the neutral wire is disconnected.
- C. to break the circuit when the live wire touches the metal case.
- D. to provide a low-resistance path when the live wire touches the metal case.

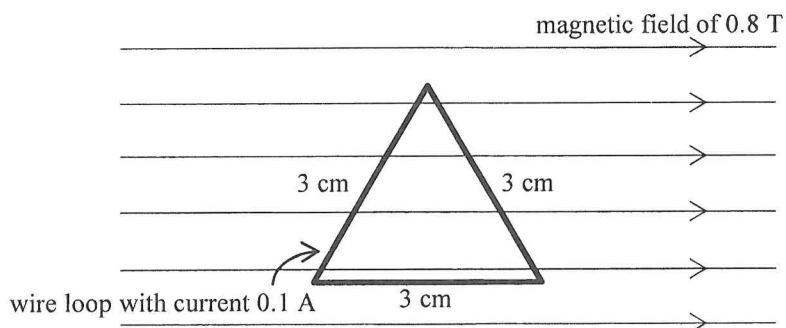
26.



If both light bulbs work at their rated values as shown, find the resistance of R .

- A. $30\ \Omega$
- B. $60\ \Omega$
- C. $90\ \Omega$
- D. $120\ \Omega$

27.



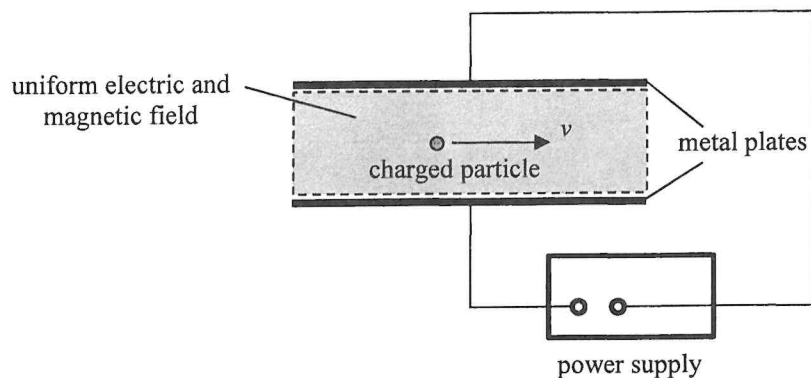
An equilateral triangular wire loop with sides of 3 cm is placed in a uniform magnetic field of 0.8 T. One of the sides of the loop is parallel to the field as shown. A current of 0.1 A flows through the loop. Find the net magnetic force acting on the loop.

- A. 0 N
- B. $2.8 \times 10^{-3}\ \text{N}$
- C. $5.6 \times 10^{-3}\ \text{N}$
- D. $7.6 \times 10^{-3}\ \text{N}$

*28. The input terminals of a transformer are connected to a 220 V mains supply. Its output power is 18 W and its output voltage is 5.0 V. If the efficiency of the transformer is 85%, what is the current drawn from the mains supply ?

- A. 0.070 A
- B. 0.082 A
- C. 0.096 A
- D. 0.55 A

*29.

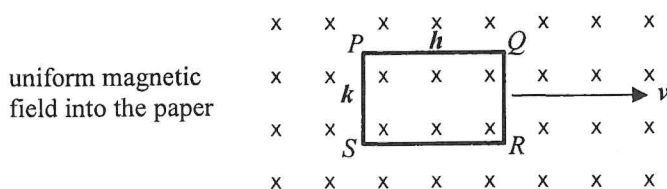


A particle with charge q travels horizontally at speed v without deflection through a region of uniform electric and magnetic field as shown. The electric field is set up by the potential difference across the two parallel metal plates. The magnetic field B is applied perpendicular to the paper. The whole set-up is in vacuum. Which of the following combinations allow the particle to travel horizontally without deflection through the region? Neglect the effects of gravity.

	charge of particle	speed of particle	magnetic field
(1)	q	$2v$	$\frac{B}{2}$
(2)	$-q$	v	B
(3)	$2q$	v	B

- A. (1) and (2) only
 B. (2) and (3) only
 C. (1) and (3) only
 D. (1), (2) and (3)

*30. A rectangular copper loop $PQRS$, with its plane perpendicular to a uniform magnetic field of flux density B , is moving with velocity v in a direction perpendicular to the field as shown. The length and width of the loop are h and k respectively.

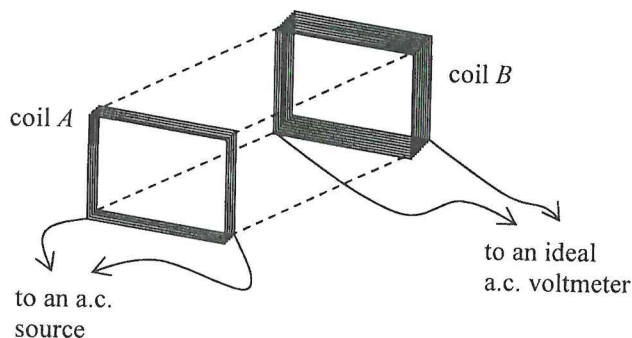


Which of the following statements is/are correct?

- (1) The potential difference across PQ is Bhv .
 (2) The potential difference across QR is Bkv .
 (3) There is no induced current flowing through the loop.

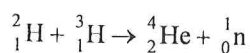
- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

- *31. Coil *A* and coil *B* are placed close and facing each other with the number of turns of coil *B* being doubled that of coil *A*. Coil *A* is connected to an a.c. source while coil *B* is connected to an ideal a.c. voltmeter as shown. The voltmeter shows a reading after the a.c. source is turned on.



Which of the following will increase the voltmeter reading ?

- (1) The voltage of the a.c. source is increased.
 - (2) A thicker wire is used in coil *B*.
 - (3) A thin and large plastic sheet is placed between the two coils and parallel to them.
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only
32. Which of the following units is used for measuring the radiation equivalent dose of a patient receiving radiotherapy treatment ?
- A. sievert (Sv)
 B. becquerel (Bq)
 C. joule (J)
 D. watt per square metre (W m^{-2})
- *33. Consider the following nuclear reaction:



Using the data below, estimate the energy released when 5 g of ${}^2_1\text{H}$ completely reacts with ${}^3_1\text{H}$ through the reaction.

mass of ${}^2_1\text{H} = 2.0136 \text{ u}$

mass of ${}^3_1\text{H} = 3.0155 \text{ u}$

mass of ${}^4_2\text{He} = 4.0015 \text{ u}$

mass of ${}^1_0\text{n} = 1.0087 \text{ u}$

- A. $3 \times 10^{-12} \text{ J}$
 B. $3 \times 10^{-10} \text{ J}$
 C. $4 \times 10^{12} \text{ J}$
 D. $2 \times 10^{18} \text{ J}$

END OF SECTION A

List of data, formulae and relationships

Data

molar gas constant	$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
acceleration due to gravity	$g = 9.81 \text{ m s}^{-2}$ (close to the Earth)
universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
speed of light in vacuum	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
charge of electron	$q_e = 1.60 \times 10^{-19} \text{ C}$
electron rest mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
atomic mass unit	$u = 1.661 \times 10^{-27} \text{ kg}$ (1 u is equivalent to 931 MeV)
astronomical unit	$\text{AU} = 1.50 \times 10^{11} \text{ m}$
light year	$\text{ly} = 9.46 \times 10^{15} \text{ m}$
parsec	$\text{pc} = 3.09 \times 10^{16} \text{ m} = 3.26 \text{ ly} = 206265 \text{ AU}$
Stefan constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$

Rectilinear motion

For uniformly accelerated motion :

$$\begin{aligned}
 v &= u + at \\
 s &= ut + \frac{1}{2}at^2 \\
 v^2 &= u^2 + 2as
 \end{aligned}$$

Mathematics

Equation of a straight line	$y = mx + c$
Arc length	$= r\theta$
Surface area of cylinder	$= 2\pi rh + 2\pi r^2$
Volume of cylinder	$= \pi r^2 h$
Surface area of sphere	$= 4\pi r^2$
Volume of sphere	$= \frac{4}{3}\pi r^3$

For small angles, $\sin \theta \approx \tan \theta \approx \theta$ (in radians)

Astronomy and Space Science $U = -\frac{GMm}{r}$ gravitational potential energy $P = \sigma AT^4$ Stefan's law $\left \frac{\Delta f}{f_0} \right \approx \frac{v}{c} \approx \left \frac{\Delta \lambda}{\lambda_0} \right $ Doppler effect	Energy and Use of Energy $E = \frac{\Phi}{A}$ illuminance $\frac{Q}{t} = \kappa \frac{A(T_H - T_C)}{d}$ rate of energy transfer by conduction $U = \frac{\kappa}{d}$ thermal transmittance U-value $P = \frac{1}{2} \rho A v^3$ maximum power by wind turbine
Atomic World $\frac{1}{2} m_e v_{\max}^2 = hf - \phi$ Einstein's photoelectric equation $E_n = -\frac{1}{n^2} \left\{ \frac{m_e q_e^4}{8h^2 \epsilon_0^2} \right\} = -\frac{13.6}{n^2} \text{ eV}$ energy level equation for hydrogen atom $\lambda = \frac{h}{p} = \frac{h}{mv}$ de Broglie formula $\theta \approx \frac{1.22\lambda}{d}$ Rayleigh criterion (resolving power)	Medical Physics $\theta \approx \frac{1.22\lambda}{d}$ Rayleigh criterion (resolving power) $\text{power} = \frac{1}{f}$ power of a lens $L = 10 \log \frac{I}{I_0}$ intensity level (dB) $Z = \rho c$ acoustic impedance $\alpha = \frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$ intensity reflection coefficient $I = I_0 e^{-\mu x}$ transmitted intensity through a medium

A1.	$E = mc \Delta T$	energy transfer during heating and cooling	D1.	$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$	Coulomb's law
A2.	$E = l \Delta m$	energy transfer during change of state	D2.	$E = \frac{Q}{4\pi\epsilon_0 r^2}$	electric field strength due to a point charge
A3.	$pV = nRT$	equation of state for an ideal gas	D3.	$E = \frac{V}{d}$	electric field between parallel plates (numerically)
A4.	$pV = \frac{1}{3} Nmc^2$	kinetic theory equation	D4.	$R = \frac{\rho l}{A}$	resistance and resistivity
A5.	$E_K = \frac{3RT}{2N_A}$	molecular kinetic energy	D5.	$R = R_1 + R_2$	resistors in series
B1.	$F = m \frac{\Delta v}{\Delta t} = \frac{\Delta p}{\Delta t}$	force	D6.	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	resistors in parallel
B2.	moment = $F \times d$	moment of a force	D7.	$P = IV = I^2 R$	power in a circuit
B3.	$E_P = mgh$	gravitational potential energy	D8.	$F = BQv \sin \theta$	force on a moving charge in a magnetic field
B4.	$E_K = \frac{1}{2} mv^2$	kinetic energy	D9.	$F = BIl \sin \theta$	force on a current-carrying conductor in a magnetic field
B5.	$P = Fv$	mechanical power	D10.	$B = \frac{\mu_0 I}{2\pi r}$	magnetic field due to a long straight wire
B6.	$a = \frac{v^2}{r} = \omega^2 r$	centripetal acceleration	D11.	$B = \frac{\mu_0 NI}{l}$	magnetic field inside a long solenoid
B7.	$F = \frac{Gm_1 m_2}{r^2}$	Newton's law of gravitation	D12.	$\mathcal{E} = N \frac{\Delta \Phi}{\Delta t}$	induced e.m.f.
C1.	$\Delta y = \frac{\lambda D}{a}$	fringe separation in double-slit interference	D13.	$\frac{V_s}{V_p} \approx \frac{N_s}{N_p}$	ratio of secondary voltage to primary voltage in a transformer
C2.	$d \sin \theta = n\lambda$	diffraction grating equation	E1.	$N = N_0 e^{-kt}$	law of radioactive decay
C3.	$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$	equation for a single lens	E2.	$t_{\frac{1}{2}} = \frac{\ln 2}{k}$	half-life and decay constant
			E3.	$A = kN$	activity and the number of undecayed nuclei
			E4.	$\Delta E = \Delta mc^2$	mass-energy relationship